

EGGROLL Scaling

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Motivation

The Problem

Definition

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called L -smooth if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|.$$

Definition

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called ρ -Hessian lipschitz if

$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq \rho\|x - y\|$$

Main Goal

$$\min_{x \in \mathbb{R}^d} f(x)$$

One problem: Even verifying if $x \in \mathbb{R}^d$ is a local minimum is NP-hard.

Stationary Points

Instead, we want to find stationary points:

Definition

A point $x \in \mathbb{R}^d$ is an ε -first order stationary point if

$$\|\nabla f(x)\| \leq \varepsilon.$$

Definition

When f is ρ -Hessian Lipschitz, a point $x \in \mathbb{R}^d$ is an ε -second-order stationary point if

$$\|\nabla f(x)\| \leq \varepsilon, \quad \nabla^2 f(x) \succeq -\sqrt{\rho\varepsilon}I.$$

Best you can (reasonably*) hope for: SOSP.

Algorithm 1 Gradient Descent

Require: Initial point $x_0 \in \mathbb{R}^d$, number of iterations T

1: **for** $k = 0, 1, \dots, T - 1$ **do**

2: $x_{k+1} \leftarrow x_k - \frac{1}{L} \nabla f(x_k)$

3: **end for**

4: **return** x_T

Theorem

In $T = O(L(f(x_0) - f^)/\varepsilon^2)$ iterations, x_T is a FOSP.*

Why Gradient Free?

- In many applications (e.g., RL), gradients are difficult or impossible to compute
- Instead, we may only have access to function evaluations
- **Question:** Can we efficiently optimize f using only zeroth-order information?

Zero-Order Methods

The main idea is to use the gradient estimator

$$\tilde{\nabla} f(x) := \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} p, \quad p \sim \mathcal{N}(0, I_d)$$

For L -smooth non-convex f , in $T = O(d/\varepsilon^2)$ iterations [3] [7]

$$\min_{t \leq T} \mathbb{E}(\|\nabla f(x_t)\|) \leq \varepsilon$$

Typically, zero-order methods take d times more iterations than first order ones

Common Problem

- The common problem with all existing ZO literature: requires $p \sim N(0, I_d)$
- But for ML, d is way too big for this to be tractable

EGGROLL

Definition (Symmetric EGGROLL Update)

With $a_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_d)$, $b_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_m)$ and $P_i = a_i b_i^\top$,

$$g_\sigma^r(X) := \frac{1}{r} \sum_{i=1}^r \frac{f(X + \sigma P_i) - f(X - \sigma P_i)}{2\sigma} P_i$$

Algorithm 2 Rank- r Symmetric EGGROLL

Require: Initial point $X_0 \in \mathbb{R}^{d \times m}$, horizon T , stepsize $\eta > 0$, smoothing radius $\sigma > 0$, rank $r \in \mathbb{N}$

for $t = 0, \dots, T - 1$ **do**

 Sample $\{a_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I_d)$ and $\{b_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I_m)$

 Update $X_{t+1} \leftarrow X_t - \eta g_\sigma^r(X_t)$

end for

return X_T

Main Theorem (informal)

Theorem

Consider running Algorithm 2. Suppose $0 < \delta \leq 1/e$. Suppose

$$\sigma = \tilde{O}\left(\frac{\varepsilon}{dm\sqrt{r}L}\right), \quad \eta = \tilde{O}\left(\frac{r^2}{dmL}\right).$$

Then in

$$T = \tilde{O}\left(\frac{dm}{r} \cdot \frac{L(f(X_0) - f^*)}{\varepsilon^2}\right)$$

iterations (with each iteration using $2r$ function evaluations), with probability at least $1 - \delta$, at least $3/4$ of the iterates are ε -first order stationary points.

Things to bound

From L -smooth you get

$$f(x + h) \leq f(x) + \langle \nabla f(x), h \rangle + \frac{L}{2} \|h\|^2$$

By using $ab \leq (a^2 + b^2)/2$ and $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ a few times,

$$\begin{aligned} f(X_{t+1}) - f(X_t) &\leq -\frac{\eta}{2r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \frac{\eta \sigma^2 L^2}{2r} \sum_{i=1}^r \|P_{t,i}\|^4 \\ &\quad + L\eta^2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{\eta^2 L^3 \sigma^2}{r} \sum_{i=1}^r \|P_{t,i}\|^6 \end{aligned}$$

To bound

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2$$

we will show it is “subgaussian enough”.

Definition (Subgaussian distribution)

Random variable X is said to be $SG(\sigma)$ if

$$\mathbb{E}[\exp(X^2/\sigma^2)] \leq 2$$

and $nSG(\sigma)$ if

$$\mathbb{E}[\exp(\|X\|^2/\sigma^2)] \leq 2$$

As an example $Z \sim \mathcal{N}(0, I)$ is $nSG(c\sqrt{d})$.

Concentration Lemma

Lemma ([5])

Let random vectors $x_1, \dots, x_n \in \mathbb{R}^d$, and corresponding filtrations $\mathcal{F}_t = \sigma(x_1, \dots, x_t)$ satisfy that

$$\mathbb{E}_{\mathcal{F}_{t-1}}[x_t] = 0, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(\|x_t\| \geq s) \leq 2e^{-\frac{s^2}{2\sigma^2}} \quad \forall s \in \mathbb{R}$$

Then there exists an absolute constant c such for any fixed $\delta > 0, \theta > 0$, with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^n x_i \right\| \leq c \cdot \theta \sum_{i=1}^n \sigma_i^2 + \frac{1}{\theta} \log\left(\frac{2d}{\delta}\right).$$

Main Theorem

Theorem

Let $\{\mathcal{F}_t\}_{t \geq -1}$ be a filtration, and let $\{a_t\}_{t \geq 0}, \{b_t\}_{t \geq 0}$ be two sequences of independent random vectors such that $a_t \sim \mathcal{N}(0, I_d)$, $b_t \sim \mathcal{N}(0, I_m)$, $a_t, b_t \in \mathcal{F}_t$ and a_t, b_t are independent of $\mathcal{F}_{t-1}$. Let $\{V_t\}_{t \geq 0}$ be a sequence of random matrices such that $V_t \in \mathcal{F}_{t-1}$. For each $\tau \geq 0$, let

$$E_\tau = \sum_{t=0}^{\tau-1} (V_t - \langle V_t, a_t b_t^\top \rangle a_t b_t^\top)$$

Then, there exists some absolute constants $C > 0, c > 0$ such that for any $\tau \in \mathbb{Z}^+$, and $0 < \delta < 1/e$, for any $\theta > 0$, with probability at least $1 - \delta$ given $\mathcal{F}_{t-1}$,

$$\|E_\tau\|_F \leq c \cdot \theta \sum_{t=0}^{\tau-1} dm \log^3 \left(\frac{C\tau m}{\delta} \right) \|V_t\|_F^2 + \frac{1}{\theta} \log \left(\frac{Cm^2 d\tau}{\delta} \right).$$

Lemma

Let X follow χ_d^2 distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d , with

$$A := \{dL_\delta^2 \leq X \leq 3d \log(C/\delta)\}$$

we have

$$\mathbb{E}[(X - d); A] = 0, \quad \mathbb{P}(A) \geq 1 - \delta$$

for some absolute constant C .

Proof of Main Theorem I

Proof

- The main trick is to use SVD of $V_t = P\Sigma Q^\top$
- By previous lemma, can define event $A_t \cap B_t$ where all $\|a_t\| \leq \sqrt{3d \log(\cdot)}$ and $|b_t^\top q_i| \leq \sqrt{3 \log(\cdot)}$ whp.
- Most crucially, $b_t^\top q_i$ are independent as q_i are orthogonal.
- Then $R_t = (V_t - \langle V_t, a_t b_t^\top \rangle a_t b_t^\top) \mathbb{1}_{A_t \cap B_t}$ is mean 0. (**)
- This is done by considering $\mathbb{E}_{\mathcal{F}_{t-1}}[Q^\top b_t b_t^\top; B_t]$.

Proof of Main Theorem II

- $\|aa^\top Vbb^\top\|_F \leq |a^\top Vb| \sqrt{9md \log(\cdot)}$ on $A_t \cap B_t$
- Given b , $a^\top Vb \sim \mathcal{N}(0, \|Vb\|^2)$
- Then $\mathbb{E}[\exp(t(a^\top Vb)^2) \mid b] \leq \exp(2t\|Vb\|^2)$
- As $Vb \sim \mathcal{N}(0, VV^\top)$, $\mathbb{E}[\exp(2t\|Vb\|^2)] \leq \exp(4t\|V\|_F^2)$
- Yields R_t is $\tilde{O}(md\|V\|_F^2)$ subgaussian.
- Utilizing lemma from before, we are done. □

The rest of the proof is in showing

$$\sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 \geq \tilde{\Omega} \left(\sum_{t=0}^{\tau-1} \|\nabla f(X_t)\| \right)$$

- Since $\mathbb{E}[|\langle \nabla f(X_t), a_t b_t^\top \rangle|^2] = \|\nabla f(X_t)\|_F^2$, know at least once every $t_f(\delta) = \frac{1}{r} \log(\cdot)$ iterations, $\frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|_F^2$
- η and σ tuned so that on intervals of length $t_f(\delta)$, $\|\nabla f(X_t)\|$ does not change by more than factor of 4
- Thus with just one t with $\frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|^2$, can replace with the sum (up to log factors)
- Which completes the proof. □

Scaling

r-Dimensional Random Projection

EGGROLL update:

$$\frac{1}{r} \left(\sum_{i=1}^r P_i P_i^\top \right) \nabla f(x)$$

Let Π_Q be the matrix which projects onto the uniformly random r -dimensional subspace Q . Then by rotational invariance

$$\mathbb{E}[\Pi_Q] = cI$$

Taking traces $\text{Tr}(P_Q) = r$ and $\text{Tr}(I) = d$, so $c = r/d$. Therefore,

$$\mathbb{E}[\|\Pi_Q g\|^2] = \mathbb{E}[\langle g, \Pi_Q g \rangle] = \langle g, \mathbb{E}[\Pi_Q] g \rangle = \frac{r}{d} \|g\|^2$$

So every iteration, we can only really capture r/d fraction of the available descent.

Towards Improvement

- Naturally, to escape $\mathbb{E}[\|\Pi_Q g\|^2] = \frac{r}{d} \|g\|^2$, need better directions.
- For L -smooth convex functions, have co-coercivity inequality

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq \frac{1}{L} \|y - x\|^2$$

When $y = x^+$ the next iterate in GD, get

$$\langle \nabla f(x^+), \nabla f(x) \rangle \geq \|\nabla f(x^+)\|^2$$

- Suggests, if we know which perturbations are well-aligned with $\nabla f(x)$, maybe they will also be with $\nabla f(x^+)$!
- Non-convex function is (nearly) convex around a local min.
- Open question: Other ideas?

Future Directions

Algorithm 3 Perturbed Gradient Descent

Require: Initial point x_0 , step size η , perturbation radius r

```
1: for  $t = 0, 1, 2, \dots$  do  
2:   if perturbation condition holds then  
3:     Sample  $\xi_t \sim \text{Unif}(B_0(r))$   
4:      $x_t \leftarrow x_t + \xi_t$   
5:   end if  
6:    $x_{t+1} \leftarrow x_t - \eta \nabla f(x_t)$   
7: end for
```

Theorem ([4])

PGD converges to SOSP in $T = O\left(\frac{L(f(x_0) - f^)}{\varepsilon^2}\right)$ iterations.*

Perturbed Gradient Descent

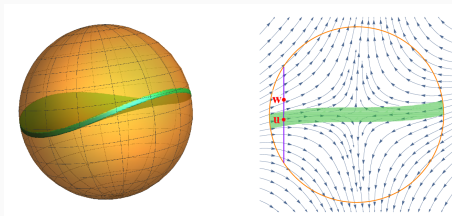
Consider quadratic $f(x) = x^\top Hx$, with $H = \text{diag}(-1, 1, \dots, 1)$

Then $x_t = x_{t-1} - \eta \nabla f(x_{t-1}) = (I - \eta H)^t x_0$

Then with $\lambda_i = \lambda_i(H)$ and $\alpha_i = e_i^\top x_0$,

$$f(x_t) - f(x_0) = \sum_{i=1}^d \lambda_i (1 - \eta \lambda_i)^{2t} \alpha_i^2 \leq -1.5^{2t} \alpha_1^2 + 0.5^{2t}$$

When $x_0 \sim \text{Unif}(B_0(1))$, whp $\alpha_1^2 = \Omega(\delta^2/d)$. Thus takes $\log(\sqrt{d}/\delta)$ steps for decrease by constant amount.



Zero-Order PGD

Algorithm 4 Perturbed Zeroth-Order Gradient Descent

Require: Initial point x_0 , horizon T , step size η , smoothing radius u , perturbation radius r , batch size m

- 1: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 2: Sample $Z_{t,1}, \dots, Z_{t,m} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$
 - 3: $g_u^{(m)}(x_t) \leftarrow \frac{1}{m} \sum_{i=1}^m \frac{f(x_t + uZ_{t,i}) - f(x_t - uZ_{t,i})}{2u} Z_{t,i}$
 - 4: Sample $Y_t \sim \mathcal{N}(0, r^2/d \cdot I_d)$
 - 5: $x_{t+1} \leftarrow x_t - \eta \left(g_u^{(m)}(x_t) + Y_t \right)$
 - 6: **end for**
 - 7: **return** x_T
-

Theorem ([8])

Perturbed zeroth-order GD converges to ε -SOSP in time

$$T = O \left(\frac{d}{m} \cdot \frac{(L + L^2) \rho^{3/2} (f(x_0) - f^*)}{\varepsilon^{5/2}} \right)$$

Algorithm 5 Nesterov Accelerated Gradient Descent

Require: Initial point $x_0 \in \mathbb{R}^d$, smoothness L , iterations T

```
1:  $y_0 \leftarrow x_0$ 
2: for  $k = 1, 2, \dots, T$  do
3:    $x_k \leftarrow y_{k-1} - \eta \nabla f(y_{k-1})$ 
4:    $y_k \leftarrow x_k + \frac{k-1}{k+2}(x_k - x_{k-1})$ 
5: end for
6: return  $x_{T+1}$ 
```

Theorem

For any $\eta \leq 1/L$, Nesterov AGD has

$$f(x_T) - f^* \leq O\left(\frac{L\|x_0 - x^*\|^2}{T^2}\right)$$

Proved using the Lyapunov $A_k(f(x_k) - f^*) + \frac{1}{2}\|z_k - x^*\|^2$

ODE View of Acceleration

In [9] they model Nesterov's AGD using a second-order ODE. By taking step size $s \rightarrow 0$ get that X satisfies

$$\ddot{X} + \frac{3}{t}\dot{X} + \nabla f(X) = 0.$$

Taking inner product with $\dot{X}$,

$$\frac{d}{dt} \left(\frac{1}{2} \|\dot{X}\|^2 + f(X) \right) = -\frac{3}{t} \|\dot{X}\|^2$$

Motivated by this, in [6] they use the Hamiltonian $E_t = f(x_t) + \frac{1}{2\eta} \|v_t\|^2$ to show perturbed AGD escapes saddle points faster than perturbed GD:

Theorem

w.p. at least $1 - \delta$, Perturbed AGD converges to SOSP in time

$$T = \tilde{O} \left(\frac{L^{1/2} \rho^{1/4} (f(x_0) - f^*)}{\epsilon^{7/4}} \right)$$

Mirror Descent

Problem: gradient descent makes no attempt to characterise lower bound on f^* . For any points $x_0, \dots, x_T$, by convexity

$$\forall u \quad f(u) \geq \frac{1}{T} \sum_{t=0}^{T-1} f(x_t) + \frac{1}{T} \sum_{t=0}^{T-1} \langle \nabla f(x_t), u - x_t \rangle$$

And so,

$$f(\bar{x}) - f(u) \leq \frac{1}{T} \sum_{t=0}^{T-1} \langle \nabla f(x_t), x_t - u \rangle =: R_T(u)$$

Considering a regularized version

$$\widetilde{R}_T(u) := \frac{1}{T} \left(-\frac{w(u)}{\alpha} + \sum_{t=0}^{T-1} \langle \nabla f(x_t), x_t - u \rangle \right)$$

By choosing $x^+ = \operatorname{argmax}_u \widetilde{R}_k(u)$, MD converges in $T = O(\rho^2/\varepsilon^2)$ iterations where ρ^2 is the average value of $\|\nabla f(x_t)\|_*^2$

MD Definitions

Definition

For closed and convex $Q \subset \mathbb{R}^d$, $w : Q \rightarrow \mathbb{R}$ is a DGF if w is 1-strongly convex wrt $\|\cdot\|$. The Bregman divergence is defined as

$$V_x(y) = w(y) - \langle \nabla w(x), y - x \rangle - w(x)$$

The properties of DGF ensures $V_x(x) = 0$ and $V_x(y) \geq \frac{1}{2}\|x - y\|^2 \geq 0$.

Examples

- $w(y) = \frac{1}{2}\|y\|_2^2$ for the ℓ^2 norm gives $V_x(y) = \frac{1}{2}\|x - y\|^2$
- When $Q = \Delta$ is the probability simplex, $w(y) = \sum_i y_i \log(y_i)$ for the ℓ^1 norm gives $V_x(y) = \sum_i y_i \log(y_i/x_i)$

Definition

The mirror descent update step with step length α

$$x^+ = \text{Mirr}_x(\alpha \nabla f(x)), \text{Mirr}_x(\xi) = \operatorname{argmin}_{y \in Q} \{V_x(y) + \langle \xi, y - x \rangle\}$$

Lemma ([2])

For all $u \in Q$,

$$\alpha(f(x) - f(u)) \leq \alpha \langle \nabla f(x), x - u \rangle \leq \frac{\alpha^2}{2} \|\nabla f(x)\|_*^2 + V_x(u) - V_{x^+}(u)$$

Theorem

Where $\Theta \geq V_{x_0}(x^*)$, by telescoping the previous lemma, MD has $f(\bar{x}) - f^* \leq \varepsilon$ in time

$$T = O\left(\frac{2\Theta L^2}{\varepsilon^2}\right)$$

The big question: Can GD and MD be combined for faster methods?

GD tells us to go one place, while MD tells us to go another. Where should we go?

$$x_{k+1} = \tau z_k + (1 - \tau)y_k$$

$$y_{k+1} = \text{Grad}(x_{k+1})$$

$$z_{k+1} = \text{Mirr}_{z_k}(\alpha \nabla f(x_{k+1}))$$

The answer: a linear combination!

Convergence

Lemma (Coupling)

When $\tau \in (0, 1)$ satisfying $\frac{1-\tau}{\tau} = \alpha L$, for all $u \in Q$,

$$\alpha \langle \nabla f(x_{k+1}), x_{k+1} - u \rangle \leq \alpha^2 L (f(y_k) - f(y_{k+1})) + V_{z_k}(u) - V_{z_{k+1}}(u)$$

Theorem

Telescoping the previous lemma,

$$\alpha T(f(\bar{x}) - f(x^*)) \leq \alpha^2 L (f(y_0) - f(y_T)) + V_{x_0}(x^*) - V_{x_T}(x^*)$$

Choosing $\alpha = \sqrt{\Theta/L\varepsilon}$, and restarting the procedure a few times, obtain ε -approx. solution in

$$T = O(\sqrt{L\Theta/\varepsilon} + \sqrt{L\Theta/2\varepsilon} + \dots) = O(\sqrt{L\Theta/\varepsilon})$$

matching Nesterov AGD.

Open Question

- Experiments show replacing $\nabla f(x)$ with the EGGROLL approximation, MD still converges.
- The proof would likely give the tools to design an accelerated ZO method by linear mixing of GD and MD.
- Problem: EGGROLL update only gives lower bound on a random r -dimensional (affine) subspace $x_t + Q_t$ each iteration.
- First idea: average projection

$$\frac{d}{r} \cdot \frac{1}{T} \sum_{t=0}^{T-1} \Pi_{x_t + Q_t} x^* \approx x^*?$$

- This has huge variance.

Variance Reduction

Consider the problem

$$\min_x f(x) := \frac{1}{n} \sum_{i=1}^n f_i(x)$$

where each f_i is L -smooth.

SGD chooses $i \sim \text{Unif}([n])$ and applies an update

$$x^+ = x - \eta \nabla f_i(x)$$

Problem: this has huge variance

$$\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x) - \nabla f(x)\|^2$$

Variance Reduction

However, if you know the full gradient at a snapshot point $\tilde{x}$, $\nabla f(\tilde{x})$, you can compute

$$v_t = \nabla f_i(x_t) - \nabla f_i(\tilde{x}) + \nabla f(\tilde{x})$$

Which has variance

$$\text{Var}(v_t) \leq O(\|x_t - \tilde{x}\|^2)$$

Then by infrequently computing the full gradient and linearly coupling three points $x_{k+1} = \tau_1 z_k + \tau_2 \tilde{x} + (1 - \tau_2 - \tau_1)y_k$,

Theorem

The algorithm in [1] produces an ε -FOSP in only

$$O\left(\frac{n^{2/3}L(f(x_0) - f(x^*))}{\varepsilon^2}\right)$$

which beats GD and SGD by $n^{1/3}$.

- For EGGROLL, maybe infrequently compute a full gradient
- Could get lower variance for gradient estimator
- By linearly coupling so the points stay close, might exploit more descent
- Potentially speeding up convergence.

Questions?

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